

Products, Unions, Algebraic Data Types

Concepts of Programming Languages
Lecture 7

Practice Problem

$\{ x : \text{int}, y : \text{int} \} \vdash (\text{fun } z \rightarrow y) x : \text{int}$

Write a derivation of the above typing judgment using the inference rules of 320Caml

Practice Problem

$$\frac{x : \tau \in \Gamma}{\Gamma \vdash x : \tau} \text{ (var)}$$

$$\frac{\textcircled{\Gamma \vdash e_1 : \tau_1 \rightarrow \tau} \quad \Gamma \vdash e_2 : \tau_1}{\Gamma \vdash e_1 e_2 : \tau} \text{ (app)}$$

$$\textcircled{\Gamma \vdash e_1 e_2 : \tau}$$

$$\frac{\Gamma, x : \tau_1 \vdash e : \tau_2}{\Gamma \vdash \text{fun } x \rightarrow e : \tau_1 \rightarrow \tau_2} \text{ (fun)}$$

$$\{x : \text{int}, y : \text{int}, z : \text{int}\} \vdash y : \text{int} \text{ (var)}$$

$$\{x : \text{int}, y : \text{int}\} \vdash \text{fun } z \rightarrow y : \text{int} \rightarrow \text{int} \text{ (fun)}$$

$$\{x : \text{int}, y : \text{int}\} \vdash x : \text{int} \text{ (var)}$$

$$\{ \mathbf{x : int, y : int} \} \vdash \mathbf{(\text{fun } z \rightarrow y) x : \text{int}} \text{ (app)}$$

Outline

- » Discuss the use of **tuples** and **records** for creating collections of data
- » Introduce **algebraic data types (ADTs)** for creating data with given "shapes"
- » Cover **parametric** and **recursive** ADTs for more general data structures

Products

Tuples

```
let point : float * float = (2.0, 3.0)  
let student : string * int = ("Franco", 244342)
```

ordered unlabeled fixed-length heterogeneous
collections of data

Primarily useful for "returning" multiple arguments

Pattern Matching on Tuples

```
let hypotenuse (p : float * float) : float =
```

```
  match p with
```

```
  | (x, y) -> sqrt (x *. x +. y *. y)
```

match e with

| true → e₁

| false → e₂

// if e then e₁ else e₂

There are no accessors for tuples

Instead we use **pattern matching**

match e with

| 0 → e₁

| 1 → e₂

| ... →

Pattern Matching in General

```
match e with  
| p -> o  
| ...
```

A **pattern** is a typed template for how a piece of data should look

A **match-expression** is a way of *deconstructing* data in OCaml

We *match* on an expression *e*, and check if the value of *e* *matches* with the pattern *p*

Note: Patterns are not Expressions

```
<expr> ::= match <expr> with  
        | <pattern> -> <expr>  
        | <pattern> -> <expr>  
        | . . .
```

Patterns are similar to expressions, but not identical

They can be wildcards, they can be variables, there's a lot of possibilities ([link](#))

Advanced Pattern Matching

```
let hypotenuse ((x, y) : float * float) : float =  
    sqrt (x *. x +. y *. y)
```

```
let hypotenuse (p : float * float) : float =  
    let (x, y) = p in  
    sqrt (x *. x +. y *. y)
```

Pattern matching can also be done implicitly in
let-expressions and function arguments

demo

And we can do all this
formally!

Tuples (Syntax Rule)

$$\langle \text{expr} \rangle ::= (\langle \text{expr} \rangle , \dots , \langle \text{expr} \rangle)$$

If $e_1, \dots, e_n$ are well-formed expressions, then

$$(e_1 , \dots , e_n)$$

is a well-formed expression

Tuple (Typing Rule)

$$\frac{\Gamma \vdash e_1 : \tau_1 \quad \Gamma \vdash e_2 : \tau_2 \quad \dots \quad \Gamma \vdash e_n : \tau_n}{\Gamma \vdash (e_1 \text{ , } \dots \text{ , } e_n) : \tau_1 * \dots * \tau_n} \text{ (tuple)}$$

If $e_1, \dots, e_n$ are of type $\tau_1, \dots, \tau_n$, respectively, in the context Γ then

$$(e_1 \text{ , } \dots \text{ , } e_n)$$

is of type $\tau_1 * \dots * \tau_n$ in the context Γ

Tuple (Semantic Rule)

$$\frac{e_1 \Downarrow v_1 \quad \dots \quad e_n \Downarrow v_n}{(e_1 \text{ , } \dots \text{ , } e_n) \Downarrow (v_1, \dots, v_n)} \text{ (tupleE)}$$

If $e_1, \dots, e_n$ evaluate to $v_1, \dots, v_n$, respectively, then

$(e_1 \text{ , } \dots \text{ , } e_n)$

evaluates to $(v_1, \dots, v_n)$

Example (Typing Derivation)

$$\begin{array}{c}
 \xrightarrow{\text{(var)}} \\
 \{x:\alpha\} \vdash x:\alpha \\
 \hline
 \{\} \vdash \text{fun } x \rightarrow x:\alpha \rightarrow \alpha \\
 \hline
 \{\} \vdash (\text{fun } x \rightarrow x, \text{fun } y \rightarrow y+y): (\alpha \rightarrow \alpha) * (\text{int} \rightarrow \text{int})
 \end{array}$$

$$\begin{array}{c}
 \xrightarrow{\text{(var)}} \\
 \{y:\text{int}\} \vdash y+y:\text{int} \\
 \hline
 \{\} \vdash \text{fun } y \rightarrow y+y:\text{int} \rightarrow \text{int} \\
 \hline
 \{\} \vdash (\text{fun } x \rightarrow x, \text{fun } y \rightarrow y+y): (\alpha \rightarrow \alpha) * (\text{int} \rightarrow \text{int})
 \end{array}$$

$$\begin{array}{c}
 \xrightarrow{\text{(var)}} \quad \xrightarrow{\text{(var)}} \\
 \{x:\text{int}\} \vdash x:\text{int} \quad \{x:\text{int}\} \vdash x:\text{int} \\
 \hline
 \{x:\text{int}\} \vdash x+x:\text{int} \quad \text{intAdd} \\
 \hline
 \{x:\text{int}\} \vdash \text{true}:\text{bool} \quad \text{(trueLit)} \\
 \hline
 \{x:\text{int}\} \vdash (x+x, \text{true}) : \text{int} * \text{bool} \quad \text{(tup)}
 \end{array}$$

Example (Semantic Derivation)

(2 + 3 , true) $\Downarrow$ (5, T)

Pair Match (Typing Rule)

$$\frac{\Gamma \vdash p : \tau_1 * \tau_2 \quad \Gamma, x : \tau_1, y : \tau_2 \vdash e : \tau}{\Gamma \vdash \text{match } p \text{ with } | x, y \rightarrow e : \tau} \text{ (pairMatch)}$$

If p is of type $\tau_1 * \tau_2$ in the context Γ and e is of type τ in the context Γ along with x begin type τ_1 and y begin type τ_2 then

$\text{match } p \text{ with } | x, y \rightarrow e$

is of type τ in the context Γ

Pair Match (Semantic Rule)

$$\frac{(e_1, e_2) \Downarrow (v_1, v_2) \quad [v_2/y][v_1/x]e \Downarrow v}{\text{match } p \text{ with } | x, y \rightarrow e \Downarrow v} \text{ (pairMatch)}$$

If (e_1, e_2) evaluates to (v_1, v_2) and e evaluates to v after substituting v_1 for x and v_2 for y , then

$\text{match } p \text{ with } | x, y \rightarrow e$

evaluates to v

Practice Problem

$\emptyset \vdash \text{fun } x \rightarrow \text{match } x \text{ with } | (a, b) \rightarrow a + b : \tau$

Determine the type τ so that the above judgment is derivable from the rules below. Also give a derivation

$$\frac{\Gamma \vdash p : \tau_1 * \tau_2 \quad \Gamma, x : \tau_1, y : \tau_2 \vdash e : \tau}{\Gamma \vdash \text{match } p \text{ with } | x, y \rightarrow e : \tau} \text{ (tuple)}$$

$$\frac{\Gamma \vdash e_1 : \text{int} \quad \Gamma \vdash e_2 : \text{int}}{\Gamma \vdash e_1 + e_2 : \text{int}} \text{ (addInt)}$$

$$\frac{\Gamma, x : \tau_1 \vdash e : \tau_2}{\Gamma \vdash \text{fun } x \rightarrow e : \tau_1 \rightarrow \tau_2} \text{ (fun)}$$

$$\frac{(v : \tau) \in \Gamma}{\Gamma \vdash v : \tau} \text{ (intLit)}$$

$\emptyset \vdash \text{fun } x \rightarrow \text{match } x \text{ with } | (a, b) \rightarrow a + b : \tau$

Solution

Records

```
type point = { x_cord : float ; y_cord : float }  
let origin : point = { x_cord = 0. ; y_cord = 0. }
```

```
type user = {  
  name : string ;  
  email : string ;  
  num_posts : int ;  
}
```

unordered labeled **fixed-length** **heterogeneous** collections
of data

They are useful for organizing large collections of data
(akin to database records)

Record Syntax (Informal)

```
type record_ty =  
  {  
    field1 : ty1;  
    field2 : ty2;  
    ...  
    fieldn : tyn;  
  }
```

```
let record_expr : record_ty =  
  {  
    field1 = expr1;  
    field2 = expr2;  
    ...  
    fieldn : exprn;  
  }
```

For a record, we have to specify the type of each field

When we construct a record, every field must have a value

Accessors

```
type point = { x_cord : float ; y_cord : float }
```

```
let dist (p : point) (q : point) =  
  let xd = p.x_cord -. q.x_cord in  
  let yd = p.y_cord -. q.y_cord in  
  sqrt (xd *. xd +. yd *. yd)
```

Records support **dot-notation**

(we can also access records by pattern matching)

Record Updates

```
let new_post u : user =  
  { u with num_posts = u.num_posts + 1 }
```

We can use **with-syntax** to update a smaller number of fields in a large record

"u with num_posts incremented, keep everything else the same"

Data in functional languages are immutable. This returns a new record with the update

Homework

Write the syntax, typing and semantic rules for records

Unions

Simple Variants

```
type os = constructor BSD | Linux | MacOS | Windows
```

A **simple variant** is a *user-defined* type for values of a fixed collection of possibilities

Type names are **lower_case** and Constructors names are **Upper_case**

Pattern Matching

```
let supported (sys : os) : bool =  
  match sys with  
  constant pattern | BSD -> false  
  wildcard pattern | _ -> true
```

We work with variants by **pattern matching**:

- » give a **pattern** that a value can **match** with
- » give an output value for each pattern

Data-Carrying Variants

```
type linux_distro = Arch | Fedora | NixOS | Ubuntu
```

```
type os
  = BSD of int * int
  | Linux of linux_distro * int
  | MacOS of int
  | Windows of int
```

Note the syntax

```
let supported (sys : os) : bool =
  match sys with
  | BSD (major, minor) -> major > 2 && minor > 3
  | _ -> true
```

Variants can carry data, which allows us to represent more complex structures

Aside: Constructor Arguments are not Tuples

```
type t = A of int * int  
let args : int * int = (2, 3)  
(* let a : t = A args *)
```

This code (uncommented) **won't type-check**

Arguments need to be passed in **directly**

Aside: Constructors are not functions

```
type t = A of int function as an argument
let apply (f : int -> t) (x : int) : t = f x
(* let x : int = apply A t *)
```

This code (uncommented) won't type check

We **cannot** partially apply constructors

Pro Tip: Named Data-Carrying Variants

```
type os
  = MacOS of { major : int; minor : int; patch : int }
  | ...

let support (sys : os) : bool =
  match sys with
  | MacOS info ->
    info.major = 10 && info.minor >= 14 && info.patch >= 1
    (* MacOS Sonoma 10.14.(1-3) *)
  | ...
```

Since we can carry *any* kind of data in a constructor, we can carry **records** to **name the parts** of our carried data

Practice Problem

```
let area (s : shape) =  
  match s with  
  | Rect r -> r.base *. r.height  
  | Triangle { sides = (a, b) ; angle } -> Float.sin angle *. a *. b  
  | Circle r -> r *. r *. Float.pi
```

*Define the variant **shape** which makes this function type-check*

What about variable
length data?

Algebraic Data Types

Recursive Data

```
type t
  = A
  | B of t
  | C of t * t
```

```
let x : t = B (C (A, B A))
```

A variant type **t** can carry data of type **t**

Why would we want to do this?

Example: Lists

```
type intlist  
  = Nil  
  | Cons of int * intlist
```

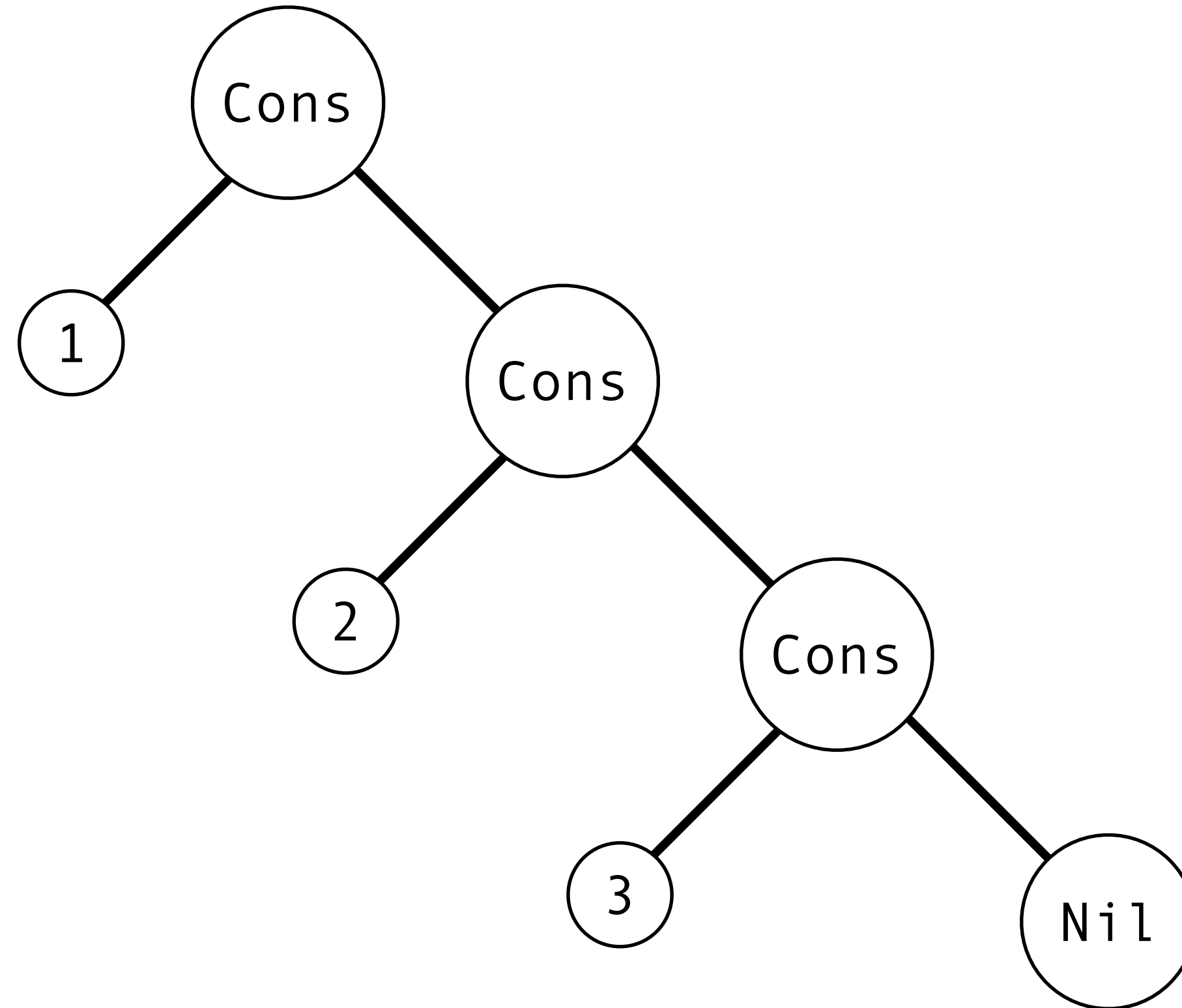
```
let example = Cons (1, Cons (2, Cons (3, Nil)))
```

The type **intlist** is available as the type of data which a constructor of **intlist** holds

We can use recursive ADTs to create variable-length data types

The Picture

```
Cons (1,  
      Cons (2,  
            Cons (3,  
                  Nil))))
```



We think of values of recursive variants as **trees** with constructors as nodes and carried data as leaves

A More Interesting Example: Expressions

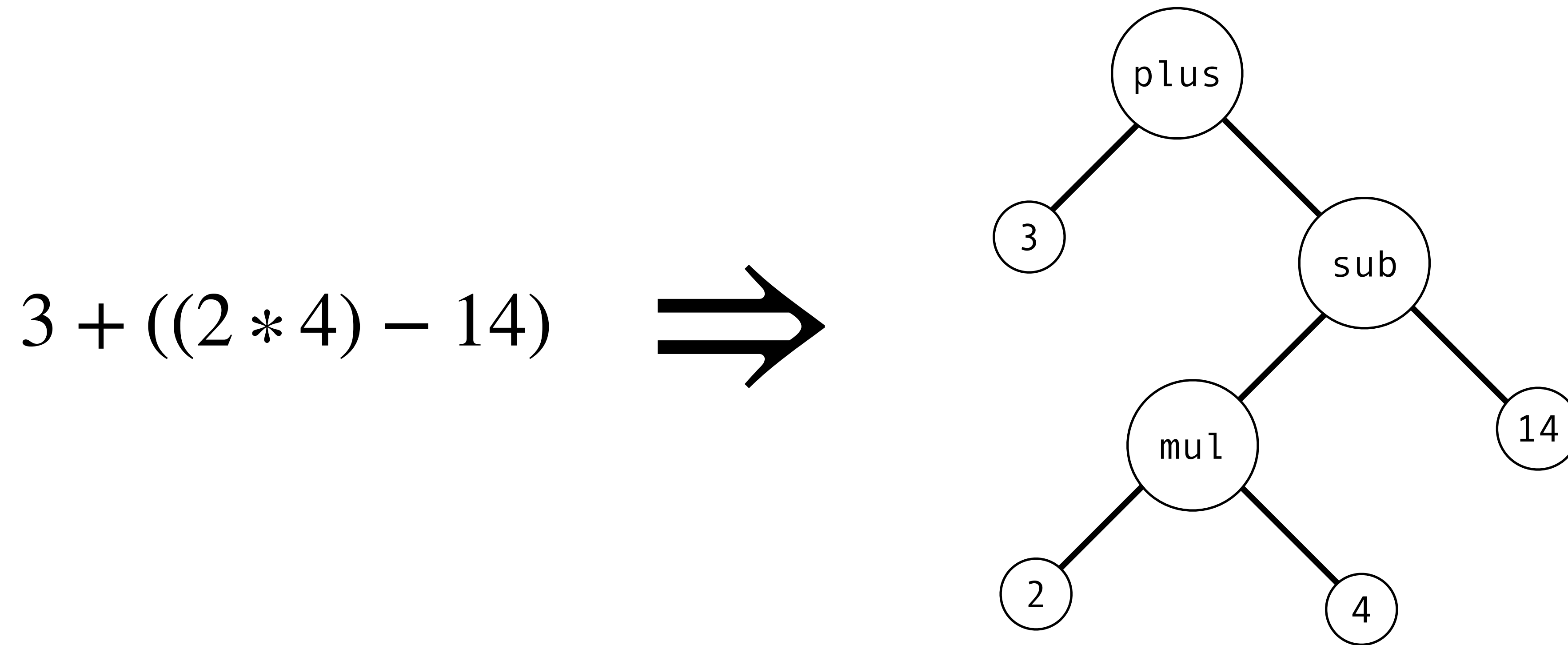
$$3 + ((2 * 4) - 14)$$

Suppose we're building a calculator...

Before we compute the value of an input, it's useful to find an **abstract representation** of the input.

This will help us separate the tasks of **evaluation** and **parsing**

A More Interesting Example: Expressions



We can represent an expression abstractly as a **tree** with operations as nodes and number values as leaves

A More Interesting Example: Expressions

```
type expr
  = Val of int
  | Add of expr * expr
  | Sub of expr * expr
  | Mul of expr * expr
```

```
let _ = Add (Val 3, Sub (Mul (Val 2, Val 4), Val 14))
```

Which means we can represent it as a recursive variant!

We 'll come back to this
next time...

Parameterized Variants

```
type type variable 'a type constructor mylist  
  = Nil  
  | Cons of 'a * 'a mylist
```

```
let e1 : int mylist = Cons (1, Cons (2, Cons (3, Nil)))  
let e2 : string mylist = Cons ("1", Cons ("2", Cons ("3", Nil)))
```

The last piece of the puzzle: variants can be type agnostic

This gives us a variant which is **parametrically polymorphic**

Recall: Parametric Polymorphism

```
let rev_tail (l : 'a list) : 'a list =  
  let rec go acc l =  
    match l with  
    | [] -> acc  
    | x :: xs -> go (x :: acc) xs  
  in go [] l
```

This allows us to write functions which can be more generally applied (reversing a list **does not depend on** what's in the list)

Useful ADTs

Recall: Lists

```
type 'a list
= []
| (::) of 'a * 'a list
```

ordered unlabeled variable-length homogeneous
collections of data

Many important operations on data can be represented as operations on lists (e.g., updating all users in a database)

Options

```
type 'a option = None | Some of 'a
```

```
let head (l : 'a list) : 'a option =  
  match l with  
  | [] -> None  
  | x :: xs -> Some x
```

Options are like boxes which *may* hold a value or may be empty

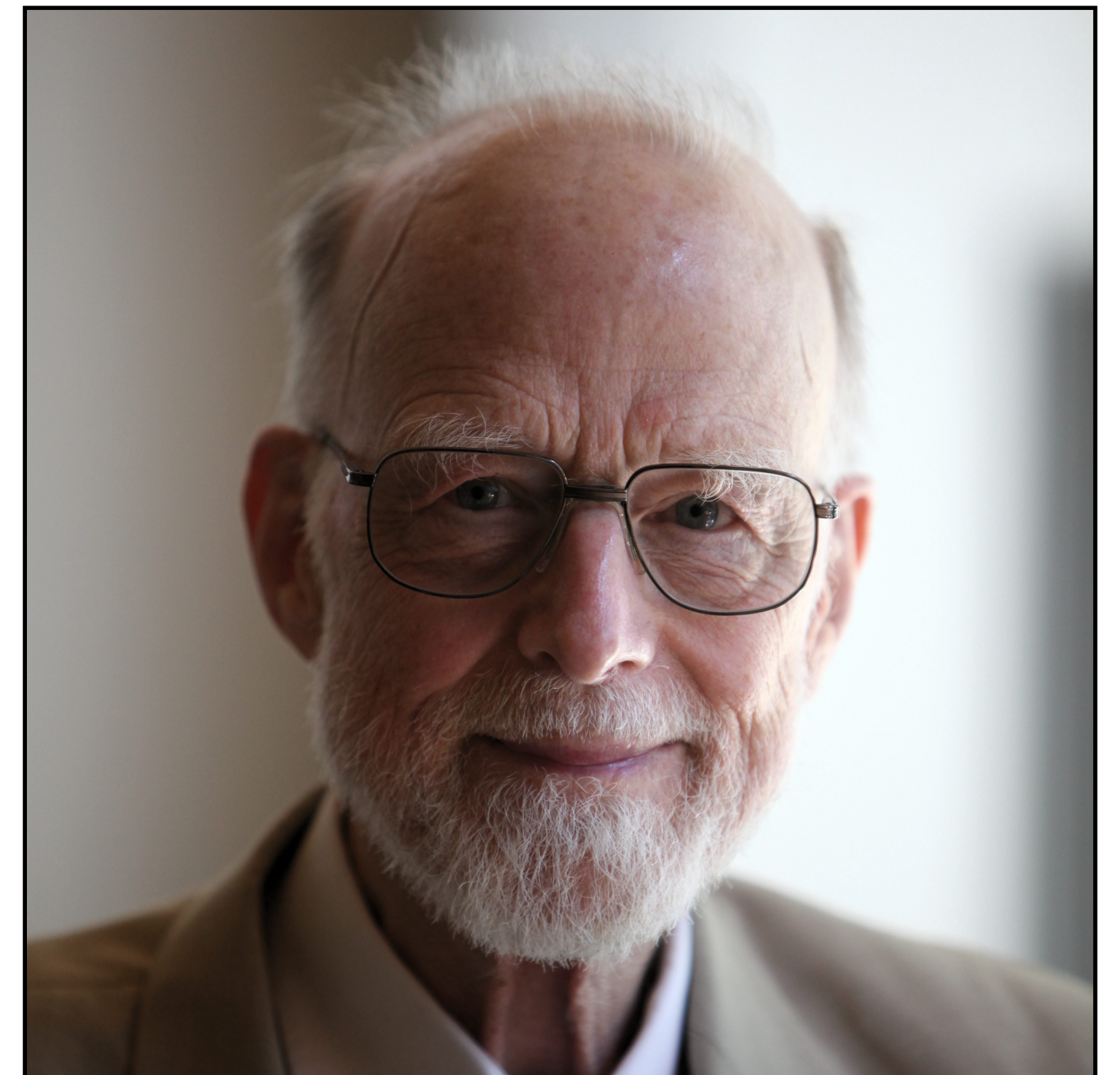
This can be useful for defining functions which **may not be total**

Aside: The Billion-Dollar Mistake

Tony Hoare calls his invention of the **null pointer** a "billion-dollar mistake"

OCaml doesn't have null pointers

I call it my billion-dollar mistake. It was the invention of the null reference in 1965. At that time, I was designing the first comprehensive type system for references in an object oriented language (ALGOL W). My goal was to ensure that all use of references should be absolutely safe, with checking performed automatically by the compiler. But I couldn't resist the temptation to put in a null reference, simply because it was so easy to implement. This has led to innumerable errors, vulnerabilities, and system crashes, which have probably caused a billion dollars of pain and damage in the last forty years.



Type-Driven Design

```
let head (l : 'a list) : 'a option =  
  match l with  
  | [] -> None  
  | x :: _ -> Some(x)
```

Types should mirror the logic of our programs.
Then we take advantage of the type checker to
verify our code (e.g., no reference to null)

Results

```
type ('a, 'e) result =  
  | Ok of 'a  
  | Error of 'e  
  
let head (l : 'a list) : ('a, string) result =  
  match l with  
  | [] -> Error "[] has no first element"  
  | x :: xs -> Ok x
```

Note the syntax

Error message

A **result** is an option with additional data in the "None" case

Aside: Built-in Variants

```
utop # #show List;;  
module List :  
  sig  
    type 'a t = 'a list = [] | (::) of 'a * 'a list  
    val length : 'a t -> int  
    val compare_lengths : 'a t -> 'b t -> int  
    val compare_length_with : 'a t -> int -> int  
    val is_empty : 'a t -> bool  
    val cons : 'a -> 'a t -> 'a t  
    val hd : 'a t -> 'a  
    ...
```

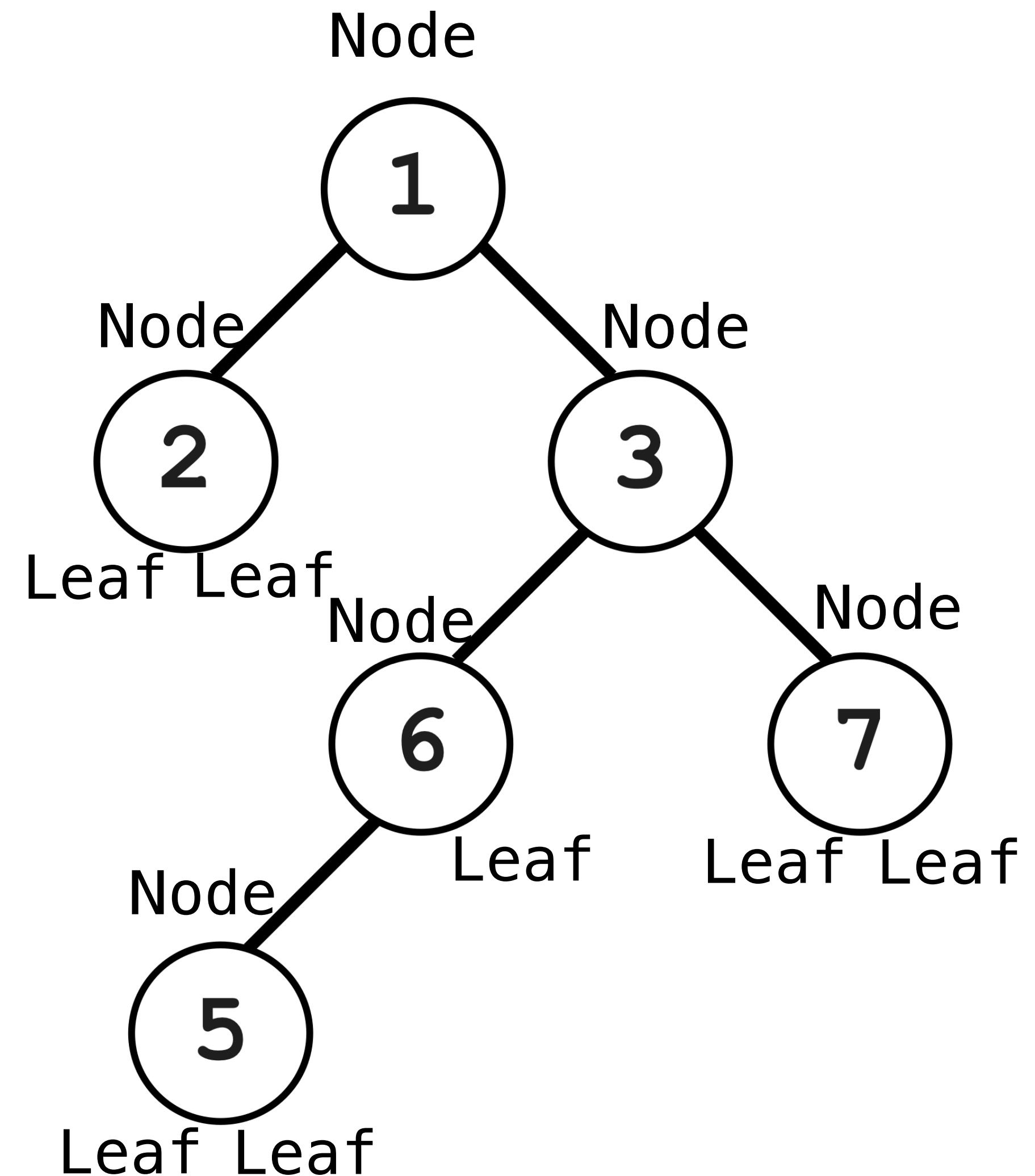
lists and optionals and results are built into OCaml

You can also use the **#show** directive in Utop to see the type signatures of functions available for lists, options and results.

Trees

```
type 'a tree =  
  | Leaf  
  | Node of 'a * 'a tree * 'a tree
```

A tree is a leaf with a value or
a node with a left or right
subtree



Summary

Tuples, records, and ADTs help us organize data and create abstract interfaces

Recursive and **parametrized** ADTs give us richer structure